

GWM-view: Gradient-weighted multi-view calibration method for machining robot positioning

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ABSTRACT

Multi-camera vision is widely used for guiding the machining robot to remove flash and burrs on complex automotive castings and forgings with arbitrary initial posture. Aiming at the problems of insufficient field of vision and regional occlusion in actual machining, a gradient-weighted multi-view calibration method (GWM-View) is proposed for the machining robot positioning based on the convergent binocular vision. Specifically, the mapping between each auxiliary camera and the main camera in the multi-view system is calculated by the inverse equation and intrinsic parameter matrix. Then, the gradient-weighted suppression algorithm is introduced to filter out the errors caused by camera angle variation. Next, the spatial coordinates of the feature points after suppression are used to correct the transformation matrix. Finally, the hand-eye calibration algorithm is implemented to transform the corrected data into the robot base coordinate system for the accurate positioning of the robot under multiple views. The experiment on the automotive engine flywheel shell indicates that the average positioning error is controlled within 1 mm under different postures. The stability and robustness of the proposed method are further improved while the positioning accuracy of the machining robot meets the requirements.

1. Introduction

The positioning accuracy of machining robots, as an important index to measure the performance of robots, restricts the precision machining process of large and complex components in the fields of aerospace, rail transit, and automotive [1–5]. Despite the visual sensors such as cameras and lasers are introduced to enhance the level of automation, but the robot positioning accuracy is easily affected by the factors of insufficient field of view and regional occlusion [6–8]. In recent years, the vision-guided robot positioning technology has broken through the visual field limitations of the traditional binocular stereo vision [9–11], structured light scanner [12–14] and laser rangefinder [15,16] by multi-view acquisition of the same object. As for the vision-guided robotic machining task, optimizing the algorithms for multi-view imaging and united calibration / correction of the existing vision system [17,18] remains a major challenge to be addressed.

In the published literatures, a series of multi-sensor fusion algorithms are proposed successively from the perspectives of temporal offset and spatial transformation between visual sensors [19], design of new laser

simulator [20], and establishment of light-space homography constraint equations [21], etc. By virtue of these means, the multi-view imaging and positioning in complex scenes are also achieved. For example, a visual servoing algorithm is proposed by Keenan et al. [22] to traverse the textureless smooth surface using the robot equipped with a monocular camera and 16 laser pointers. Li et al. [23] performed the robotic eye-in-hand calibration of the line laser sensor based on the 3D reconstructed point clouds. The laser scanner in the work of Zhang et al. [24] and Zhou et al. [25] guided the robot to complete pose identification and autonomous obstacle avoidance by the rapidly united calibration of multiple sensors. In terms of the imaging of visual system, Eckenhoff et al. [26] designed a real-time multi-camera estimator, which seamlessly fuses the multimodal information of any uncalibrated camera. Karanam et al. [27] proposed a new dynamic fusion algorithm to achieve the three-dimensional spatial positioning of the target object in a method of non-contact multi-view imaging fusion. Zhu et al. [28] proved the feasibility of 3D stereo imaging by using the natural randomness of scattering media under large parallax and multiple viewing angles.

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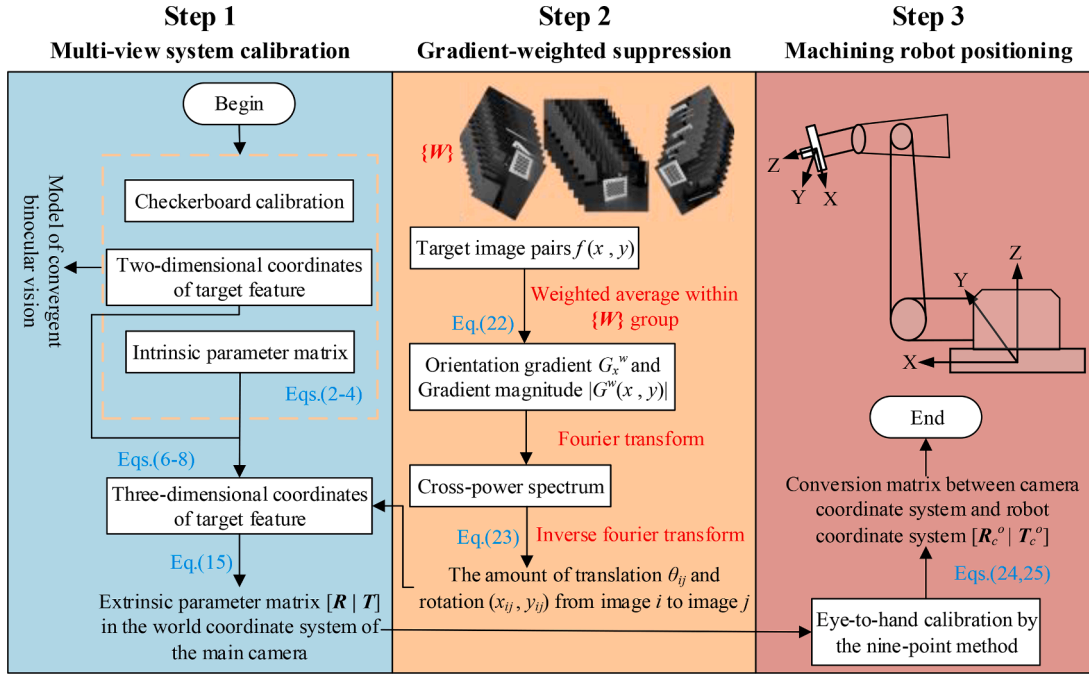


Fig. 1. Flow chart of the multi-view system guided machining robot positioning.

In addition, extensive research has been done in the calibration and correction of visual system. Kim et al. [29] proposed an algorithm of multi-view 3D display correction to compensate dislocation, and the image distortion caused by uncalibrated image was solved with a mathematical model of multi-view display. Munaro et al. [30] proposed an algorithm for camera pose correction based on a global network by integrating algorithms of optimizing RGB-D camera calibration and visual tracking, which achieved the target tracking and accurate positioning within a range of multiple views. The work by Li et al. [31] and Cheng et al. [32] calibrated and corrected the cameras in different postures based on the multi-view calibration template and high-resolution 3D stereo model, respectively. Relating to the vision-guided robotic machining, the vision-based calibration and correction methods can also address the problem of robot path planning.

By utilizing the approaches of global measurement [33–35] and real-time measured point clouds [36–38], the robotic machining trajectory can be generated and corrected simultaneously. Although the multi-view imaging is characterized by the large range, the problems of complex reconstruction of 3D model and poor stability of camera calibration algorithm also exist.

In light of the aforementioned limitations, this paper proposes a GWM-View calibration method based on the convergent binocular camera, which aims to enhance the positioning accuracy of the robot for the subsequent machining of the complex component with arbitrary initial posture. The algorithm process consists of three steps as illustrated in Fig. 1. **Step 1:** Multi-view system calibration. By calibrating the checkerboard images in the public field of view, the intrinsic parameter matrix of each camera in the visual system and the three-dimensional

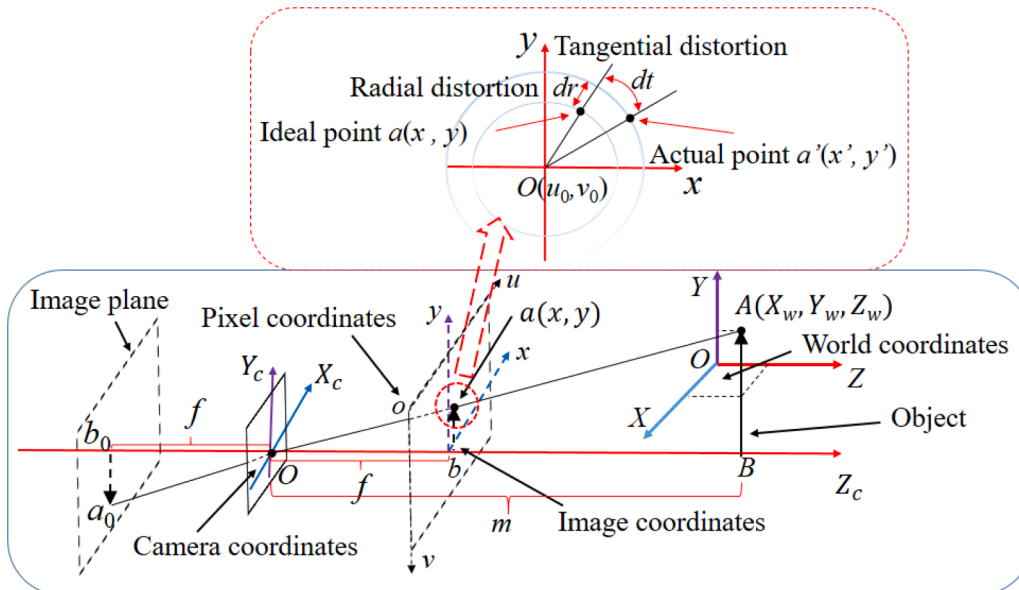


Fig. 2. Pinhole imaging and the distortion correction model.

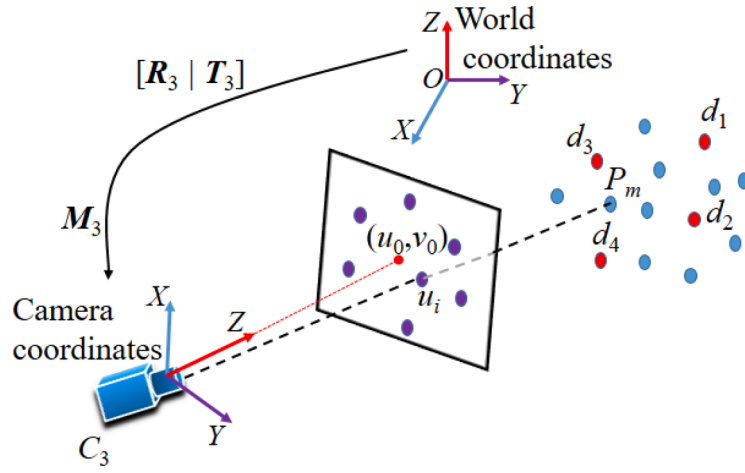


Fig. 5. Schematic diagram of the extrinsic parameter matrix reverse calibration.

y_r) and the three-dimensional coordinates of the space point $A(X_w, Y_w, Z_w)$.

Taking the left camera O_l as the main camera, the origin of camera coordinate system is regarded as the origin of world coordinate system. The pose relation of the camera coordinate system is expressed as Eq. (3) by the transformation matrix $\mathbf{M}_l^r$, that is, the relationship between the left and right cameras is directly established.

$$\begin{bmatrix} X_r \\ Y_r \\ Z_r \\ 1 \end{bmatrix} = \mathbf{M}_l^r \begin{bmatrix} X_l \\ Y_l \\ Z_l \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_l^r | \mathbf{T}_l^r \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X_l \\ Y_l \\ Z_l \\ 1 \end{bmatrix} = \begin{bmatrix} r_1 & r_2 & r_3 & t_x \\ r_4 & r_5 & r_6 & t_y \\ r_7 & r_8 & r_9 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_l \\ Y_l \\ Z_l \\ 1 \end{bmatrix} \quad (3)$$

where $\mathbf{R}_l^r = \begin{bmatrix} r_1 & r_2 & r_3 \\ r_4 & r_5 & r_6 \\ r_7 & r_8 & r_9 \end{bmatrix}$ and $\mathbf{T}_l^r = (t_x, t_y, t_z)^T$ represent the extrinsic parameter rotation matrix and translation vector of the right camera relative to the left (main) camera.

Both the two-dimensional coordinates (x_l, y_l) and (x_r, y_r) of the image coordinate system and the internal parameter matrices $\mathbf{K}_l = \begin{bmatrix} f_l & 0 & u_l \\ 0 & f_l & v_l \\ 0 & 0 & 1 \end{bmatrix}$ and $\mathbf{K}_r = \begin{bmatrix} f_r & 0 & u_r \\ 0 & f_r & v_r \\ 0 & 0 & 1 \end{bmatrix}$ of the left and right cameras are combined into Eqs. (1) and (3). The relational matrix between the binocular cameras is further expressed as Eq. (4).

$$\begin{bmatrix} x_r \\ y_r \\ 1 \end{bmatrix} = \frac{1}{Z_r} \begin{bmatrix} f_r r_1 & f_r r_2 & f_r r_3 & f_r t_x \\ f_r r_4 & f_r r_5 & f_r r_6 & f_r t_y \\ r_7 & r_8 & r_9 & t_z \end{bmatrix} \begin{bmatrix} \frac{Z_l x_l}{f_l} \\ \frac{Z_l y_l}{f_l} \\ \frac{Z_l}{f_l} \\ 1 \end{bmatrix} \quad (4)$$

2.2. Multi-view imaging model and calibration

Fig. 4 shows a multi-view imaging model with three cameras based on the convergent binocular vision. $\mathbf{R}_i^l$ and $\mathbf{T}_i^l$ represent the rotation matrix and translation vector between cameras C_i and C_j . $\mathbf{M}_i$ represents the intrinsic parameter matrix of the camera system.

The spatial pose of each camera in the same world coordinate system is obtained by solving the mapping between the three-dimensional spatial coordinates and the two-dimensional pixel coordinates of the

feature points in common area under multiple views. Assuming that the world coordinate system coincides with the coordinate system of the main camera C_1 , all the camera matrices $\mathbf{H}_i$ relative to the world coordinate system in the multi-view system are expressed as Eq. (5).

$$\begin{cases} \mathbf{H}_1 = \mathbf{M}_1 [\mathbf{R}_1 | \mathbf{0}_{3 \times 1}] \\ \mathbf{H}_2 = \mathbf{M}_2 [\mathbf{R}_{12} | \mathbf{T}_{12}] \\ \dots \\ \mathbf{H}_i = \mathbf{M}_i \left[\prod_{j=2}^i \mathbf{R}_{(i-1)j} \sum_{j=2}^i \left(\prod_{k=2}^{j-1} \mathbf{R}_{(i-1)k} \right) \mathbf{T}_{j(i-1)} \right] \end{cases} \quad (5)$$

where $I \geq 2$ represents the number of cameras (I is taken as 3 in the paper), $\mathbf{R}$ is the 3×3 identity matrix, $\mathbf{R}_{(i-1)i}$ and $\mathbf{T}_{j(i-1)}$ represent the rotation matrix and translation vector between adjacent cameras, and $\mathbf{M}_i$ is the i -th intrinsic parameter matrix. Take the auxiliary camera C_3 for example. The intrinsic parameter matrix $\mathbf{M}_3$ can be obtained by calibration in advance, and the extrinsic parameter matrix $[\mathbf{R}_3 | \mathbf{T}_3]$ needs to be solved.

Fig. 5 shows the reverse calibration of the extrinsic camera parameter matrix. The three-dimensional coordinates of any feature point d_n ($n=4$ and not in the same plane) in the space are obtained from the binocular camera model and calibration [41]. P_m ($m=1, 2, 3, \dots, m$) is the arbitrary spatial reference point of the camera.

Eq. (6) shows the transformation between the spatial feature point $d_n^w(x_w, y_w, z_w)^T$ and the camera mapping point $d_n^c(x_c, y_c, z_c)^T$. The reference point P_m is weighted by the feature point d_n .

$$\begin{cases} P_m^w = \sum_{n=1}^4 \beta_{mn} d_n^w, \sum_{n=1}^4 \beta_{mn} = 1 \\ d_n^c = [\mathbf{R}_3 | \mathbf{T}_3] \begin{bmatrix} d_n^w \\ 1 \end{bmatrix} \end{cases} \Rightarrow P_m^c = \sum_{n=1}^4 \beta_{mn} d_n^c \quad (6)$$

where P_m^w and P_m^c represent the coordinates of reference points in space and camera, respectively, and β_{mn} represents the homogeneous barycenter coordinate, which is determined by the feature point.

Given the camera intrinsic parameter matrix $\mathbf{M}_3 = \begin{bmatrix} f_x & 0 & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix}$,

the homogeneous relationship between the corresponding points of two-dimensional pixel coordinate and three-dimensional camera coordinate is established as shown in Eq. (7).

$$s_m \begin{bmatrix} u_m \\ v_m \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \sum_{n=1}^4 \beta_{mn} \begin{bmatrix} x_n^c \\ y_n^c \\ z_n^c \end{bmatrix}, s_m = \sum_{n=1}^4 \beta_{mn} z_n^c \quad (7)$$

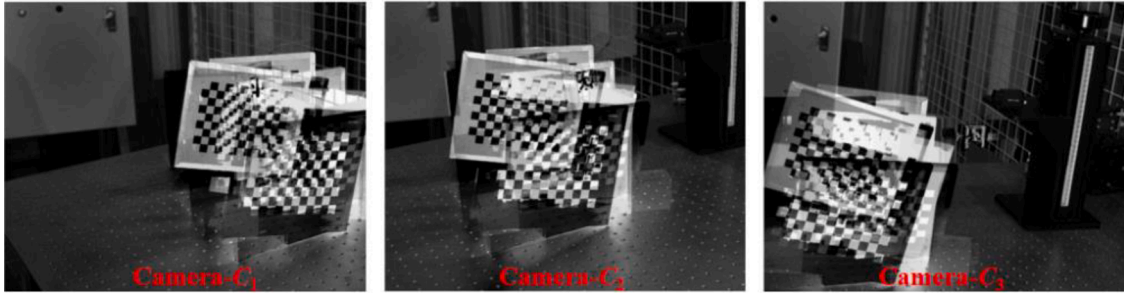


Fig. 6. Checkerboard sequence images for calibration.

By expanding and simplifying Eq. (7), the system of equations is shown in Eq. (8).

$$\begin{cases} \sum_{n=1}^4 (\beta_{mn} f_x x_n^c + \beta_{mn} (u_0 - u_i) z_n^c) = 0 \\ \sum_{n=1}^4 (\beta_{mn} f_y y_n^c + \beta_{mn} (v_0 - v_i) z_n^c) = 0 \end{cases} \quad (8)$$

where f_x, f_y, u_0 and v_0 represent the intrinsic camera parameters, and (u_i, v_i) represents the two-dimensional pixel coordinate of reference points, all of which are known quantities. $[x_n^c, y_n^c, z_n^c]^T$ represents the three-dimensional camera coordinate of feature points, which is an unknown quantity.

Eq. (8) is expressed in the form of linear equation as $D \cdot X = 0$, where D represents the known quantity matrix of $2m \times 12$ and X represents the unknown quantity matrix of 12×1 . The right singular vector V_i is further converted into Eq. (9).

$$D \cdot X = 0 \Rightarrow \begin{cases} D^T D = V_i \left(\sum \sum \right) V_i^T \\ X = \sum_{i=1}^n \varphi_i V_i \end{cases} \quad (9)$$

In combination with the space coordinate $d_n^w(x_w, y_w, z_w)^T$ of the four feature points, and the equal distance between feature points in the world coordinate system and camera coordinate system, the coefficient φ_i is determined by Eq. (10).

$$\left\| \left(\varphi_1 V_1^{[n]} + \varphi_2 V_2^{[n]} + \varphi_3 V_3^{[n]} + \varphi_4 V_4^{[n]} \right) - \left(\sum_{i=1}^4 \varphi_i V_i^{[m]} \right) \right\|^2 = \|d_n^w - d_m^w\|^2 \quad (10)$$

where $V^{[n]}$ represents the vector which is composed of the n -th feature point coordinates in V . Eq. (10) is expressed as $L \cdot \varphi = 0$, and $\varphi = [\varphi_{11}, \varphi_{12}, \varphi_{13}, \varphi_{14}, \varphi_{22}, \varphi_{23}, \varphi_{24}, \varphi_{33}, \varphi_{34}, \varphi_{44}]^T$. Similarly, by virtue of the equal distance between two feature points, the objective function is established by Eq. (11), where m and n represent the feature points and satisfy $1 \leq m < n \leq 4$.

$$E_{m,n}(\varphi) = \sum_{(m,n), s.t. m < n} (\|d_m^c - d_n^c\|^2 - \|d_m^w - d_n^w\|^2) \quad (11)$$

By solving Eq. (11), the Jacobian matrix $J_{m,n}$ of $E_{m,n}(\varphi)$ relative to φ is calculated as Eq. (12), where K_i represents the matrix of the i -th right singular vector.

$$J_{m,n} = \partial[E_{m,n}(\varphi)] / \partial \varphi = \begin{bmatrix} 2\varphi_1 K_1^T K_1 + 2\varphi_2 K_1^T K_2 + 2\varphi_3 K_1^T K_3 + 2\varphi_4 K_1^T K_4 \\ 2\varphi_1 K_2^T K_2 + 2\varphi_2 K_2^T K_2 + 2\varphi_3 K_2^T K_3 + 2\varphi_4 K_2^T K_4 \\ 2\varphi_1 K_3^T K_3 + 2\varphi_2 K_3^T K_3 + 2\varphi_3 K_3^T K_3 + 2\varphi_4 K_3^T K_4 \\ 2\varphi_1 K_4^T K_4 + 2\varphi_2 K_4^T K_4 + 2\varphi_3 K_4^T K_4 + 2\varphi_4 K_4^T K_4 \end{bmatrix}^T \quad (12)$$

Further, the Jacobian matrices of four feature points are given by Eq. (13), where ω represents the residual matrix.

$$\begin{cases} J = [J_{1,2} J_{1,3} J_{1,4} J_{2,3} J_{2,4} J_{3,4}] \\ \omega = [E_{1,2}(\varphi) E_{1,3}(\varphi) E_{1,4}(\varphi) E_{2,3}(\varphi) E_{2,4}(\varphi) E_{3,4}(\varphi)] \end{cases} \quad (13)$$

Both Eqs. (12) and (13) are combined to solve the incremental equation of φ , and the optimal solution is obtained through several iterations. As shown in Eq. (14), μ is the coefficient of φ and can be used for iterative updating.

$$J^T J \mu \varphi = -J^T \omega \Rightarrow \varphi = \varphi + \mu \varphi \quad (14)$$

After obtaining φ by the least square optimization algorithm, both P_m^w and P_c^c are known. The reference point after removing the centroid in the world coordinate system and camera coordinate system is represented in matrix, and the rotation matrix and translation vector $[R_3 | T_3]$ are obtained inversely by the singular value decomposition. As shown in Eq. (15), the centroid coordinates of the reference point are $P_c^c = \frac{1}{n} \sum_{i=1}^n P_i^c$ and $P_c^w = \frac{1}{n} \sum_{i=1}^n P_i^w$.

$$\begin{cases} q^c = \begin{bmatrix} (P_1^c)^T \dots (P_c^c)^T \\ \vdots \\ (P_n^c)^T \dots (P_c^c)^T \end{bmatrix} \\ q^w = \begin{bmatrix} (P_1^w)^T \dots (P_c^w)^T \\ \vdots \\ (P_n^w)^T \dots (P_c^w)^T \end{bmatrix} \end{cases} \Rightarrow Q = (q^c)^T q^w \Rightarrow [U \Sigma V] \\ = SVD(Q) \Rightarrow \begin{cases} R_3 = UV^T \\ T_3 = P_c^c - R_3 P_c^w \end{cases} \quad (15)$$

3. Gradient-weighted suppression algorithm

After calibrating the multi-view system, the feature points of the target image are matched to obtain the actual three-dimensional coordinates of the main camera in the world coordinate system, which is used for guiding the robot positioning. In order to increase the field of view in the vision system, a large offset angle between the auxiliary cameras and the main camera may be generated, which easily produces matching errors and then affects the final positioning accuracy. In this paper, the target image is transformed from the spatial domain to the frequency domain, and a gradient-weighted suppression function is constructed by using the interval gradient magnitude and phase correlation to correct the transformation matrix between cameras in the vision system [42,43]. Fig. 6 shows the checkerboard sequence image for calibration, which is collected by cameras under the same background. The collected three images are taken as the first group of target images to solve the image gradient.

Assuming that the original image is represented by the function $f(x,$

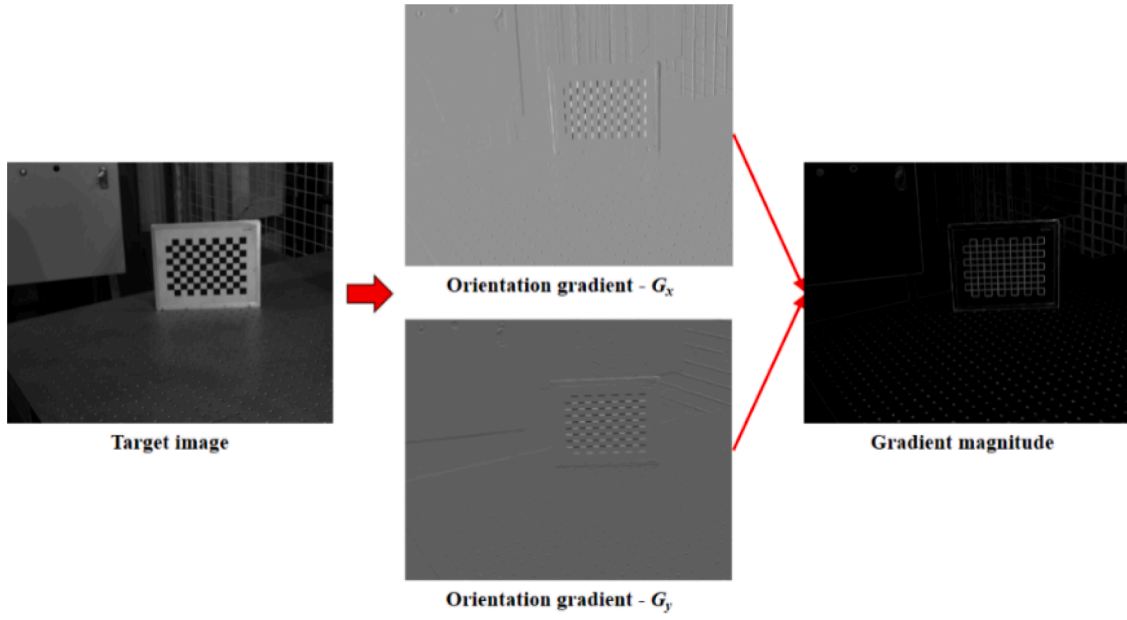


Fig. 7. Schematic diagram of the orientation gradients and gradient magnitude.

y). The gradient of a point $A(x,y)$ in the image is represented by vector as shown in Eq. (16), where $G[f(x,y)]$ is the image gradient, including the change rate of gray level at point $A(x,y)$ along the coordinate axis, and points to the direction with the greatest change.

$$G[f(x,y)] = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} \quad (16)$$

The backward difference method is then used to approximately calculate the gradient magnitude of the image as shown in Eq. (17), where $|G(x,y)|$ represents the gradient magnitude or the modulus of the gradient, $|f(x+1,y)-f(x,y)|$ represents the gradient G_x of the image along the horizontal direction, and $|f(x,y+1)-f(x,y)|$ represents the gradient G_y of the image along the vertical direction.

$$\begin{aligned} |G(x,y)| &= \left((\partial f(x,y)/\partial x)^2 + (\partial f(x,y)/\partial y)^2 \right)^{\frac{1}{2}} \\ &= ((f(x+1,y) - f(x,y))^2 + (f(x,y+1) - f(x,y))^2)^{\frac{1}{2}} \\ &\approx |f(x+1,y) - f(x,y)| + |f(x,y+1) - f(x,y)| \end{aligned} \quad (17)$$

Fig. 7 shows the calculation results of the orientation gradients and gradient magnitude of the target image. The high-pass filter composed of the horizontal filter $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ and vertical filter $\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$ is used to process the target image, and the orientation gradients G_x and G_y are obtained. The obtained gradient magnitude is further used for input information of the image processing in frequency domain.

Fourier transform is used to convert the target image from space domain to frequency domain, which can improve the matching accuracy and algorithm stability owing to better signal-to-noise ratio and phase

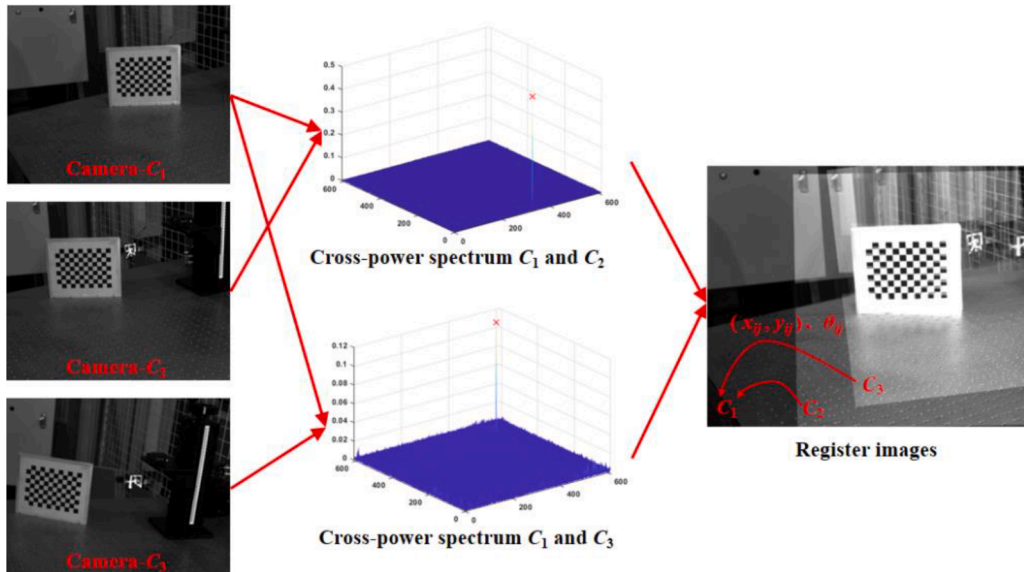


Fig. 8. Diagram of the image registration.

characteristics. Assuming that the digital image $f(x,y)$ is an $M \times N$ matrix composed of discrete gray information, the corresponding spectrum after Fourier transform is shown in Eq. (18), where $u=0,1,2,\dots,M-1$ and $v=0,1,2,\dots,N-1$ represent the frequency variables.

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)} \quad (18)$$

Similarly, the inverse Fourier transform from frequency domain to space domain is expressed in Eq. (19).

$$f(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)} \quad (19)$$

Assuming that the target checkerboard image $f'(x',y')$ after removing the background is obtained by the translation (x_0,y_0) and rotation θ_0 of $f(x,y)$, the rectangular coordinates of rotation are converted to the polar coordinate system and can be expressed as $x=r\cos\theta$, $y=r\sin\theta$, $u=\omega\cos\varphi$ and $v=\omega\sin\varphi$. The spectrum before and after the change by Eq. (18) is expressed as Eq. (20).

$$\begin{cases} f(x-x_0, y-y_0) \\ f(r, \theta + \theta_0) \end{cases} \Rightarrow \begin{cases} F'(u', v') = e^{-j2\pi(ux_0+vy_0)} F(u, v) \\ F'(\omega', \varphi) = e^{-j2\pi(r\omega(\cos\varphi\cos\theta_0+\sin\varphi\sin\theta_0))} F(\omega, \varphi) \end{cases} \quad (20)$$

According to the Fourier transform, the transform magnitude is not affected by the image rotation and translation. By calculating the normalized cross-power spectrum between the two images, the phase correlation of the two images in polar and rectangular coordinate systems is obtained accordingly. In Eq. (21), $F^*(u,v)$ and $F^*(\omega,\varphi)$ are the conjugate complex number of $F(u,v)$ and $F(\omega,\varphi)$, respectively.

$$Cp(u,v) = \begin{cases} \frac{F'(u', v') F^*(u, v)}{|F'(u', v') F^*(u, v)|} = e^{j2\pi(ux_0+vy_0)} \\ \frac{F'(\omega', \varphi) F^*(\omega, \varphi)}{|F'(\omega', \varphi) F^*(\omega, \varphi)|} = e^{j2\pi(r\omega(\cos\varphi\cos\theta_0+\sin\varphi\sin\theta_0))} \end{cases} \quad (21)$$

Combining with Eq. (19), $Cp^{-1}(u,v)=\delta([r, \theta+\theta_0]+[x-x_0, y-y_0])$ is obtained by the inverse Fourier transform of the cross-power spectrum $Cp(u,v)$. The coordinates at the maximum magnitude are the rotation angle θ_0 and the translation parameter (x_0,y_0) between the two images. Taking the first group of three target images as the registration object, the results are shown in Fig. 8, where (x_{ij},y_{ij}) and θ_{ij} represent the translation amount and rotation angle from image i to image j , $1 \leq j < i \leq 3$.

Fig. 8 shows that the images collected by the multi-view system under different views are solved by the gradient magnitude and phase correlation, and the regional matching of cameras C_2 and C_3 to camera C_1 is also achieved. Assuming that there are W groups of checkerboard sequence images, the orientation gradient-weighted G_x^w and G_y^w are constructed in Eq. (22).

$$\begin{aligned} |G^w(x,y)| &= \frac{\sum_{w=1}^W ((f(x+1,y)-f(x,y))^2 + (f(x,y+1)-f(x,y))^2)^{\frac{1}{2}}}{W} \\ &\approx \left(\sum_{w=1}^W |f(x+1,y)-f(x,y)| + \sum_{w=1}^W |f(x,y+1)-f(x,y)| \right) / W = (G_x^w + G_y^w) / W \end{aligned} \quad (22)$$

where $|G^w(x,y)|$ represents the weighted average gradient magnitude and $w=1,2,3,\dots,W$ represents the number of image groups. Eqs. (19), (21) and (22) are used simultaneously to calculate the weighted gradient magnitude by the fast Fourier transform and the inverse Fourier transform of the cross-power spectrum. Eq. (23) shows the conversion between multi-view imaging planes after gradient-weighted suppression.

Table 1

The pseudo-code of the GWM-View calibration method.

Algorithm 1 GWM-View calibration method
1: Input: Image of the target artifact I_{ij} , intrinsic camera parameter matrix $\mathbf{M}_n$ ($n=3$), the number of checkerboard image pairs W , physical distance between corners L
2: for $i = 1, 2, 3 \dots$ do
3: Solve the image coordinates of feature points (x, y) by Eq. (2)
4: for $j = 1, 2, 3 \dots$ do
5: if $n \leq 2$ then
6: Solve the conversion matrix between cameras $[\mathbf{R} \mid \mathbf{T}] \leftarrow (\mathbf{M}_n, L)$ by Eq. (4)
7: else $n > 2$ then
8: if $W > n$ then
9: Solve the weighted average gradient magnitude $|G^w(x,y)|$ by Eq. (22)
10: Solve the inverse Fourier transform of cross-power spectrum $(Cp^w)^{-1}$ by Eq. (23)
11: Solve the phase correlation between images $[(x_{ij}, y_{ij}), \theta_{ij}]$ by Eq. (23)
12: Solve the 3D coordinates of feature points $(P_m^w, P_m^w) \leftarrow (\mathbf{M}_n, (x,y))$ by Eq. (6)
13: Solve the corrected conversion matrix between cameras $[\mathbf{R} \mid \mathbf{T}]$ by Eq. (15)
14: end if
15: end if
16: end for
17: end for

$$\begin{cases} FG(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \left| \frac{G_x^w + G_y^w}{W} \right| e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)} \\ Cp^w(u,v) = \frac{FG'(u,v) FG^*(u,v)}{|FG'(u,v) FG^*(u,v)|} = e^{j2\pi(ux_0+vy_0)} \end{cases} + \begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ r^2 = x^2 + y^2 \\ \tan\theta = \frac{y}{x}, x \neq 0 \end{cases}$$

$$\Rightarrow \begin{cases} FG'(u,v) = e^{-j2\pi(ux_0+vy_0)} FG(u\cos\theta + v\sin\theta, -u\sin\theta + v\cos\theta) \\ f'(x', y') = f \left(\frac{\sum_{w=1}^W (x\cos\theta_{ij} + y\sin\theta_{ij} - x_{ij})}{W}, \frac{\sum_{w=1}^W (-x\sin\theta_{ij} + y\cos\theta_{ij} - y_{ij})}{W} \right) \end{cases} \quad (23)$$

where $M \times N$ represents the pixel matrix composed of M rows and N columns, (x,y) and (x',y') represent the pixel coordinates of the imaging plane of the auxiliary and main cameras, (x_{ij},y_{ij}) and θ_{ij} represent the translation amount and rotation angle from image i to image j .

The multi-view imaging model is introduced into the extrinsic parameter matrix of the auxiliary camera relative to the main camera, which is used for correcting the conversion matrix. The pseudo-code of the GWM-View calibration method is listed in Table 1.

4. Multi-vision guided machining robot positioning

The three-dimensional coordinates of the feature points under the main camera are converted into the robot base coordinate system by using the nine-point method of the “hand-eye” calibration algorithm [4, 6, 44, 45]. Fig. 9 shows the calibration diagram of “eye-to-hand”, which mainly consists of two modules: the coordinate system of multi-view system and the coordinate system of robot.

Given the transformation between the robot TCP coordinate system and the base coordinate system, the auxiliary camera coordinate systems in multi-view system are converted into the main camera coordinate system. The calibration process of nine-point method is shown in Eq. (24).

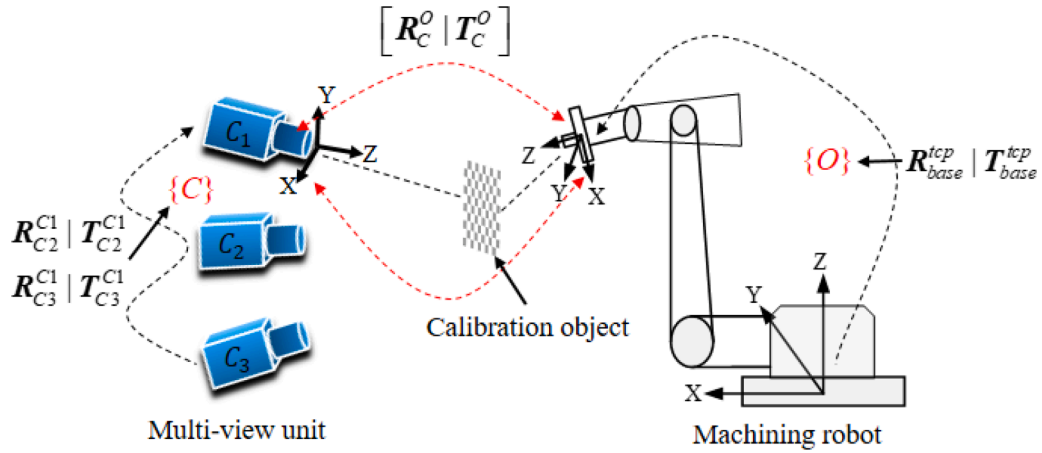


Fig. 9. Calibration diagram of the "eye-to-hand".

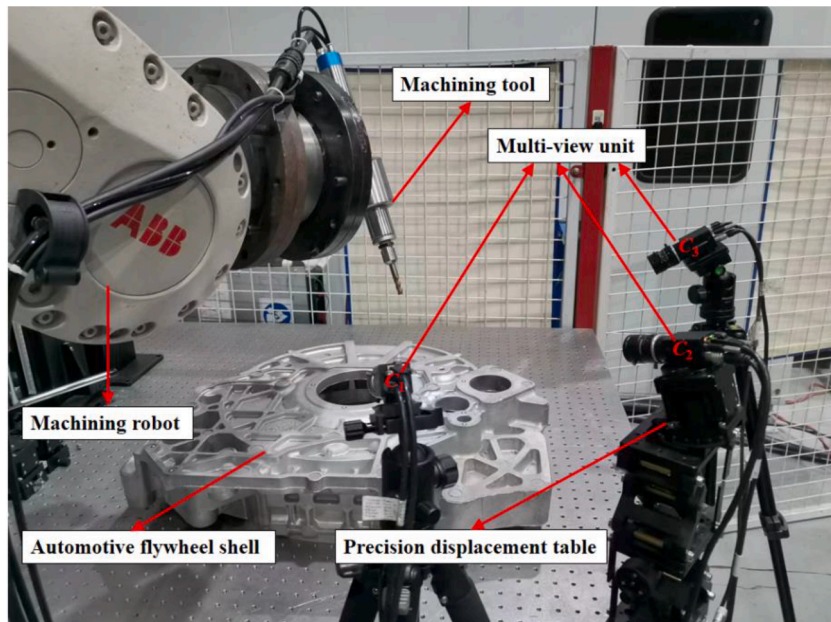


Fig. 10. Experimental platform of the multi-vision guided machining robot positioning.

$$\begin{bmatrix} x^o \\ y^o \\ z^o \\ 1 \end{bmatrix} = \mathbf{R}_C^O \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} + \mathbf{T}_C^O \Rightarrow \begin{cases} x^o = a_0x + b_0y + c_0 \\ y^o = a_1x + b_1y + c_1 \\ z^o = a_2x + b_2y + c_2 \end{cases} \quad (24)$$

where (x^o, y^o, z^o) is the robot base coordinate, (x, y) is the camera pixel coordinate, $\mathbf{R}_C^O$ and $\mathbf{T}_C^O$ represent the rotation matrix and translation vector from camera coordinate system to the robot coordinate system, a , b and c are the parameters obtained by the operation of matrix $[\mathbf{R}_C^O | \mathbf{T}_C^O]$, respectively. In combination with the calibration principle of the multi-view system, the relationship matrix between the main camera coordinate system and the robot base coordinate system is established as shown in Eq. (25).

$$\begin{bmatrix} x^o \\ y^o \\ z^o \\ 1 \end{bmatrix} = \begin{bmatrix} f_l & 0 & u_l \\ 0 & f_l & v_l \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_C^O | \mathbf{T}_C^O \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X_l \\ Y_l \\ Z_l \\ 1 \end{bmatrix} = \begin{bmatrix} f_l a_0 & f_l b_0 & f_l c_0 & f_l t_x \\ f_l a_1 & f_l b_1 & f_l c_1 & f_l t_y \\ a_2 & b_2 & c_2 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_l \\ Y_l \\ Z_l \\ 1 \end{bmatrix} \quad (25)$$

The converted three-dimensional coordinate data is input into the robot simulation software to correct the corresponding markers in the

3D model. After obtaining the workpiece posture relative to the robot base coordinate in the real environment, the robot simulation software is used to program the machining path offline, and then the robot positioning and machining operation is realized under the guidance of vision.

5. Experimental verification

5.1. Experimental setup

In order to verify the effectiveness and feasibility of the proposed method, the experimental platform of the multi-vision guided machining robot positioning is built as shown in Fig. 10. The experimental platform includes one industrial robot (ABB 6700-200/2.60) with repeated positioning accuracy of 0.05mm, three industrial cameras (MV-CS060-10GM, resolution: 3027×2048) with 25mm focal length lens, central console (GPU/Nvidia GeForce RTX 3060(12GB)), checkerboard, automotive engine flywheel shell, optical precision displacement table, motorized spindle and positioning probe. The programming and modeling software used are VisualStudio-2019, MATLAB-R2019b, CATIA-V5 and RobotStudio-6.08. Based on the convergent

Table 2

Calibration results of the convergent binocular camera.

	Camera-C ₁ and Camera-C ₂		Camera-C ₂ and Camera-C ₃	
	Left	Right	Left	Right
M	[2557.1, 0, 620.6;	[2553.1, 0, 639.5;	[2572.7, 0, 606.2;	[2575.6, 0, 508.1;
	0, 2555.6,	0, 2551.4,	0, 2595.9,	0, 2589.9,
	482.7;	511.3;	551.9;	571.1;
	0, 0, 1]	0, 0, 1]	0, 0, 1]	0, 0, 1]
R	[0.999849, -0.017067, -0.003308;		[0.983411, -0.181169, 0.008948;	
	0.017070, 0.999854, 0.000927;		0.180097, 0.981091, 0.070885;	
	0.003292, -0.000983, 0.999994]		-0.021621, -0.068098, 0.997444]	
T	[181.222, -1.454, 6.315] ^T		[186.542, -25.121, 31.599] ^T	

Table 3

Transformation matrix results before and after correction.

		Camera-C ₁	Camera-C ₂	Camera-C ₃
Before	M	[2557.99, 0, 639.14;	[2557.327, 0, 619.825;	[2558.609, 0, 640.886;
		0, 2555.62, 498.00;	0, 2557.009, 483.389;	0, 2558.219, 508.469;
		0, 0, 1]	0, 0, 1]	0, 0, 1]
	R	[1.00, 0.00, 0.00;	[0.9989, -0.0162, -0.0030;	[0.9786, -0.1939, 0.0428;
		0.00, 1.00, 0.00;	0.0156, 0.9995, -0.0046;	0.1927, 0.9765, 0.0565;
		0.00, 0.00, 1.00]	0.0031, 0.0046, 0.9998]	-0.0520, -0.0487, 0.9871]
	T	[0.00, 0.00, 0.00] ^T	[181.308, -1.639, 8.316] ^T	[355.515, -45.610, 32.592] ^T
After	M	[2557.99, 0, 639.14;	[2557.327, 0, 619.825;	[2558.609, 0, 640.886;
		0, 2555.62, 498.00;	0, 2557.009, 483.389;	0, 2558.219, 508.469;
		0, 0, 1]	0, 0, 1]	0, 0, 1]
	R	[1.00, 0.00, 0.00;	[0.9998, -0.0167, -0.0029;	[0.9799, -0.1948, 0.0422;
		0.00, 1.00, 0.00;	0.0167, 0.9998, -0.0045;	0.1923, 0.9796, 0.0576;
		0.00, 0.00, 1.00]	0.0031, 0.0045, 0.9999]	-0.0526, -0.0483, 0.9974]
	T	[0.00, 0.00, 0.00] ^T	[181.335, -1.351, 7.769] ^T	[365.233, -28.114, 50.578] ^T

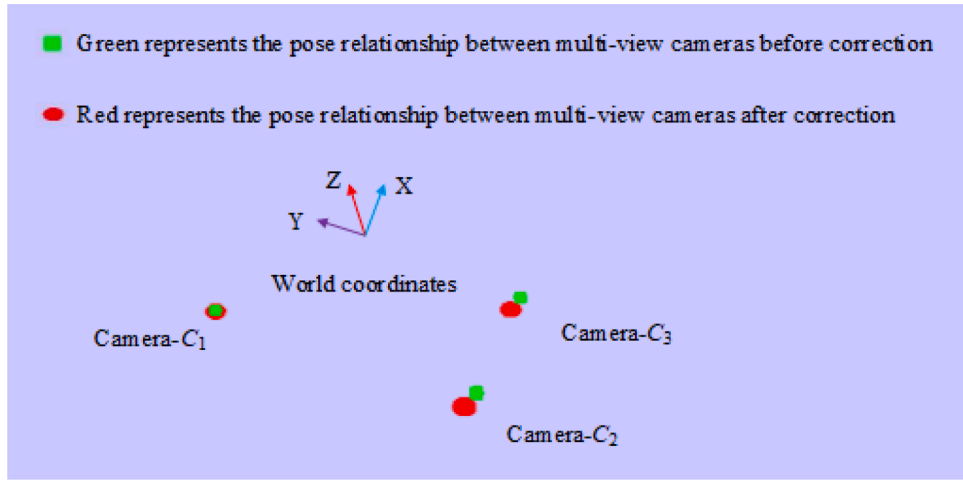
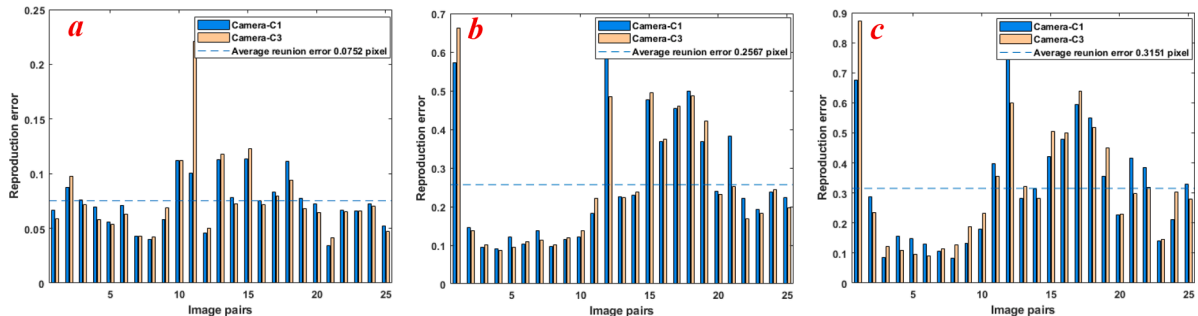
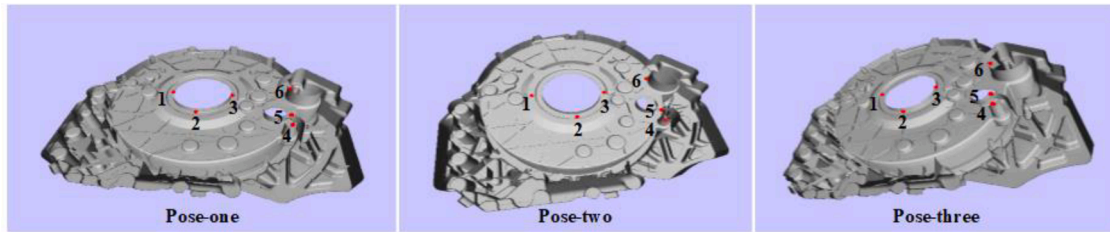
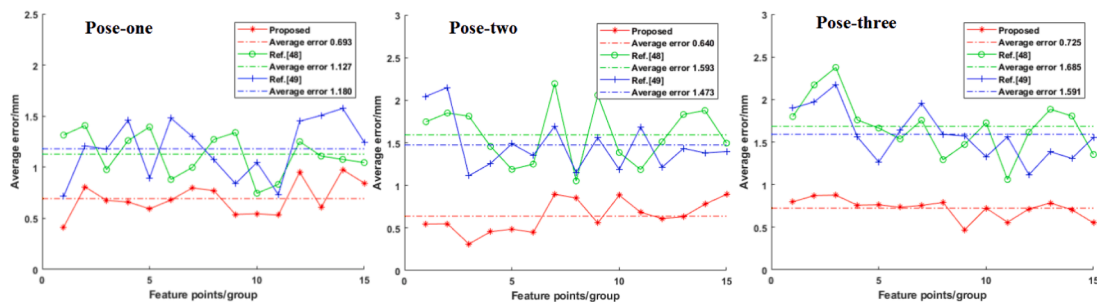
**Fig. 11.** Schematic diagram of the pose relationship between multi-view cameras before (green) and after (red) correction.**Fig. 12.** Comparison of the average reproduction errors by the proposed algorithm (a) and existing algorithms in [46] (b), [47] (c).

Table 4

Three-dimensional coordinates of the feature points of automotive flywheel shell in robot base coordinate system.

Automobile flywheel shell		1	2	3	4	5	6
Pose-one	X	1899.505	1798.360	1729.314	1457.841	1435.615	1435.357
	Y	2116.765	2028.239	2117.286	1838.435	1883.901	1912.051
	Z	1225.052	1224.795	1225.138	1208.095	1207.912	1207.434
Pose-two	X	1869.216	1762.331	1753.106	1481.523	1459.217	1459.788
	Y	2143.807	2054.092	2093.475	1813.976	1859.133	1889.115
	Z	1225.185	1225.427	1225.398	1207.707	1207.338	1208.023
Pose-three	X	1895.926	1794.853	1725.908	1454.427	1431.234	1430.986
	Y	2064.538	1976.283	2065.691	1787.769	1833.429	1861.852
	Z	1276.274	1278.566	1282.275	1269.436	1268.972	1270.133

**Fig. 13.** Schematic diagram of the three-dimensional coordinates of feature points.**Fig. 14.** Comparison of the average positioning errors of automotive flywheel shell feature points under three poses.

matrix, and T represents the 3×1 extrinsic camera parameter translation vector.

Both the rotation matrix and translation vector of the auxiliary cameras C_2 and C_3 relative to the main camera C_1 in the multi-view system are obtained according to the coordinates of the reference points in the world coordinate system and camera coordinate system. The gradient-weighted suppression function is constructed by using the interval gradient magnitude and phase correlation, and the obtained transformation matrix is corrected to enhance the three-dimensional coordinate accuracy and robustness of the algorithm for the workpiece surface feature points. The transformation matrix before and after correction is listed in Table 3, and the pose relationship between multi-view cameras before and after correction is shown in Fig. 11.

After correcting the transformation matrix in the multi-view system by the gradient-weighted suppression function, the checkerboard calibration experiment is further conducted for the auxiliary camera C_3 and the main camera C_1 . Both the concentric circle collinearity-based calibration algorithm in [46] and the binocular plane target-based calibration algorithm in [47] are chosen to compare the reproduction errors. As shown in Fig. 12, the average reproduction error by the proposed method is only 0.0752 pixel compared with the other two methods, and the accuracy is raised by nearly an order of magnitude.

Taking the automotive engine flywheel shell as the experimental object, the workpiece position and pose are changed arbitrarily in the field of multi-view system. Six groups of feature points are randomly selected (the distance between any two groups of feature points is from 50mm to 350mm), and the three-dimensional coordinates are listed in

Table 4. The three-dimensional coordinate data of the workpiece surface feature points in the real environment are input into the simulation software as shown in Fig. 13. The robot can be guided to realize the positioning and subsequent machining of the workpiece with arbitrary initial posture by correcting the pose of original 3D model.

In order to further verify the vision-guided robot positioning accuracy under multiple views, the three-dimensional coordinate distance between different feature points on the surface of automotive flywheel shell is calculated and then compared with the measured values. Both the parameter and error monitoring-based algorithm in [48] and the precise ellipse detection and grating conversion-based algorithm in [49] are chosen to compare the average positioning errors. Fig. 14 shows that the average positioning error of the proposed method can be controlled within 1mm for the complex workpiece with arbitrary initial posture. Compared with the existing algorithms, the proposed algorithm also shows better positioning accuracy and robustness.

6. Conclusions

Based on the convergent binocular vision, a GWM-View calibration method is proposed to enhance the machining robot positioning accuracy in this paper. By building an experimental platform and comparing the proposed method with the existing camera calibration algorithms, the following conclusions are achieved:

- (1) The problems of insufficient field of view and regional occlusion facing the vision-guided machining robot positioning can be

effectively solved by introducing a multi-view imaging model. By virtue of this model, the conversion matrices between the auxiliary cameras and the main camera are inversely solved while calibrating the intrinsic parameter matrices of each camera.

- (2) The errors caused by excessive camera offset can be filtered out by constructing a gradient-weighted suppression function when calculating the interval gradient magnitude and phase correlation in multi-view system. The stability and robustness of the calibration method are further enhanced by correcting the conversion matrices between the auxiliary cameras and the main camera.
- (3) The average reproduction error of the proposed method is 0.0752 pixels, which is nearly an order of magnitude smaller than the existing calibration algorithms. The average positioning error of the complex automotive flywheel shell with arbitrary initial posture can be controlled within 1mm, which effectively ensures the positioning accuracy of the vision-guided machining robot.

CRedit authorship contribution statement

Hongdi Liu: Conceptualization, Methodology, Software, Writing – original draft. **Jiahao Fu:** Data curation. **Minqi He:** Validation. **Lin Hua:** Supervision. **Dahu Zhu:** Supervision, Resources, Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare no conflict of interest.

Data availability

Data will be made available on request.

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